

# Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

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| [Website](#)

| [Linkedin](#)

| [Google Scholar](#)

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## Education

**Embry-Riddle Aeronautical University**  
M.S. in Aviation  
Specialization in Aviation Safety Systems

Daytona Beach, FL  
Aug 2025–Present  
GPA 4.0/4.0

**Embry-Riddle Aeronautical University**  
B.S. in Aeronautical Science  
Minor in Air Transportation

Daytona Beach, FL  
May 2023  
GPA 3.85/4.0

## Research Interests

Flight Safety Analytics; ADS-B Analysis; Approach Stability Assessment; Terminal Airspace Operations

## Research Experience

### Graduate Researcher

School of Graduate Studies, College of Aviation

Advisor: Dr. João Souza Dias Garcia

Sep 2025–Present

- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- ML-based aircraft fuel burn prediction using ADS-B and ACARS data
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Multi-layer evaluation of AAM disaster response and government-operated public use cases

### Graduate Researcher

Applied Aerospace Research & Development Lab

Advisor: Dr. Ryan Wallace

Feb 2026–Present

- Nationwide low-altitude traffic density modeling
- ADS-B-based airspace analysis and emerging aviation operations

## Publications & Presentations

### Publications & Proceedings

[1] Yoo, T., & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. American Institute of Aeronautics and Astronautics AVIATION Forum.

### Presentations

- Characterizing Air Traffic Density Using Nationwide ADS-B Data. (Poster). Embry-Riddle Aeronautical University Discovery Day, 2026.
- Data-Driven Framework for Assessing Wind Impacts on Approach Stability in Flight Training. (Poster). BCAAS Third Annual Safety Research Symposium, 2026
- Approach Stability Assessment in Flight Training Using ADS-B Data. (Poster). Embry-Riddle Aeronautical University Student Research Symposium, 2025.

## Teaching Experience

### Graduate Teaching Assistant

Supervisor: Dr. Dahai Liu

Embry-Riddle Aeronautical University

Dec 2025–Present

Teaching support for:

- Statistical Analysis for Aviation/Aerospace
- Research Methods in Aviation/Aerospace
- Graduate Capstone Research Project

## **Flight Instructor**

Instructional areas include:

- Private Pilot Certificate
- Instrument Rating
- Commercial Pilot Certificate
- Certified Flight Instructor
- Certified Flight Instructor-Instrument
- Flight Review and Instrument Proficiency Check
- Complex Endorsement

Aviators Academy  
Nov 2023–July 2025

## **Honors and Awards**

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|----------------------------------------------------------------------------------------------|---------------|
| Best Poster Award, Embry-Riddle Aeronautical University                                      | April 2026    |
| 1st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University              | November 2025 |
| Magna Cum Laude, Embry-Riddle Aeronautical University                                        | May 2023      |
| Deans List (6 semesters), B.S. in Aeronautical Science, Embry-Riddle Aeronautical University |               |

## **Certifications and Licenses**

Commercial Pilot – ASEL, AMEL, Instrument Airplane  
Flight Instructor – ASEL, AMEL, Instrument Airplane  
Ground Instructor – Advanced, Instrument  
Remote Pilot – sUAS

## **Programming Skills**

Python (pandas, traffic, scikit-learn, Matplotlib)  
SPSS (PROCESS macro)

## **Professional Memberships**

American Institute of Aeronautics and Astronautics  
University Aviation Association  
Royal Aeronautical Society

## **Leadership & Service**

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|-----------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Secretary</b><br>Korean Student Association                                          | Embry-Riddle Aeronautical University<br>2019–2020 |
| <b>Administrative Specialist; Squad Commander</b><br>Republic of Korea Military Service | Korea Defense Communication Command<br>2017–2019  |
| <b>Vice Captain</b><br>Table Tennis Club                                                | University of North Dakota<br>2016–2017           |